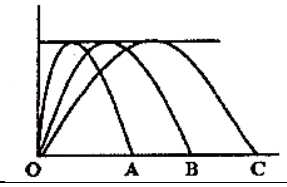
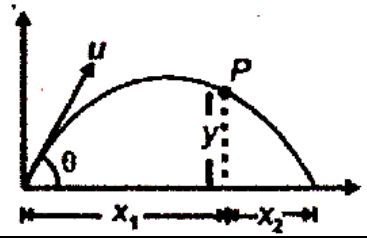
Question	A football player throws a ball with a velocity of 50metre/sec at an angle 30 degrees from the horizontal. The ball remains in the air for $(g = 10m/s^2)$
Type	multiple choice
Option	2.5sec
Option	1.25sec
Option	5sec
Option	0.625sec
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	The horizontal range is four times the maximum height attained by a projectile. The angle of projection is
Type	multiple choice
Option	90°
Option	60°
Option	45°
Option	30°
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A particle is projected from ground with speed $80m/s$ at an angle 30° with horizontal from ground. The magnitude of average velocity of particle in time interval $t = 2\text{s}$ to $t = 6\text{s}$ is [Taking $g = 10m/s^2$]
Type	multiple choice
Option	$40\sqrt{2}m/s$
Option	$40m/s$
Option	Zero
Option	$40\sqrt{3}m/s$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Two objects are thrown up at angles of 45° and 60° respectively, with the horizontal. If both objects attain same vertical height, then the ratio of magnitude of velocities with which these are projected is
Type	multiple choice
Option	$\frac{\sqrt{2}}{3}$
Option	$\frac{\sqrt{3}}{2}$
Option	$\frac{2}{3}$
Option	$\frac{\sqrt{2}}{2}$
Answer	4

Solution	
Positive Marks	4
Negative Marks	1
Question	At what point of a projectile motion acceleration and velocity are perpendicular to each other
Type	multiple choice
Option	At the point of projection
Option	At the point of drop
Option	At the topmost point
Option	Any where in between the point of projection and topmost point
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A shell is fired vertically upwards with a velocity v_1 from the deck of a ship moving with a speed v_2. A person on the shore observes the motion of the shell as a parabola. Its horizontal range is given by
Type	multiple choice
Option	$\frac{2v_1v_2}{g}$
Option	$\frac{2v_1^2v_2}{g}$
Option	$\frac{2v_1v_2^2}{g}$
Option	$\frac{2v_1^2v_2^2}{g}$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A boy can throw a stone up to maximum height of $10m$. The maximum horizontal distance that the boy can throw the same stone up to will be
Type	multiple choice
Option	$20\sqrt{2}m$
Option	$10m$
Option	$10\sqrt{2}m$
Option	$20m$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	The maximum range of a gun of horizontal terrain is $16km$. If $g = 10ms^{-2}$, then muzzle velocity of a shell must be
Type	multiple choice
Option	$160ms^{-1}$
Option	$200\sqrt{2}ms^{-1}$
Option	$400ms^{-1}$
Option	$800ms^{-1}$

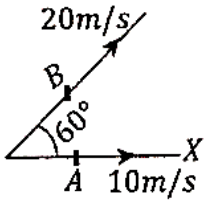
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Three projectiles A, B and C are thrown from the same point in the same plane. Their trajectories are shown in the figure. Which of the following statement is true? 
Type	multiple choice
Option	The time of flight is the same for all the three
Option	The launch speed is largest for particle C
Option	The horizontal velocity component is largest for all particle C
Option	All of the above
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	For an object from ground with speed u horizontal range is two times the maximum height attained by it. The horizontal range of object is
Type	multiple choice
Option	$\frac{2u^2}{3g}$
Option	$\frac{3u^2}{4g}$
Option	$\frac{u^2}{2g}$
Option	$\frac{4u^2}{5g}$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	The equation of a projectile is $y = 16x - 4x^2$. Its horizontal range is
Type	multiple choice
Option	4
Option	$\frac{1}{4}$
Option	16
Option	$\frac{1}{16}$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	When a particle is projected at some angle to the horizontal, it has a range R and time of flight t_1. If the same particle is projected with same speed at some other

	angle to have the same range, its time of flight is t_2. Then angle of projection will be
Type	multiple choice
Option	$\theta = \sin^{-1}(\frac{t_1}{t_2})$
Option	$\theta = \cos^{-1}(\frac{t_1}{t_2})$
Option	$\theta = \tan^{-1}(\frac{t_1}{t_2})$
Option	$\theta = \tan^{-1}(1+\frac{t_1}{t_2})$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	In the given figure for a projectile 
Type	
Option	$y = \left[\frac{x_1 x_2}{x_1 - x_2}\right]\tan\theta$
Option	$y = \left[\frac{x_1 x_2}{x_1 + x_2}\right]\tan\theta$
Option	$y = \left[\frac{2x_1 x_2}{x_1 + x_2}\right]\cos\theta$
Option	$y = \left[\frac{2x_1 x_2}{x_1 + x_2}\right]\tan\theta$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	A particle is projected with a velocity u making an angle θ with the horizontal. At any instant, its velocity v is at right angle to its initial velocity u; then v is
Type	multiple choice
Option	u cos θ
Option	u tan θ
Option	u cot θ
Option	u sec θ
Answer	3
Solution	
Positive Marks	4
Negative Marks	1

Question	The equation of a projectile is $y=\sqrt{3}x-\frac{gx^2}{2}$ the angle of projection is:
Type	multiple choice
Option	30°
Option	60°
Option	45°
Option	None
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	A helicopter is flying horizontally at an altitude of 2 km with a speed of 100 ms $^{-1}$. A packet is dropped from it. The horizontal distance between the point where the packet is dropped and the point where it hits the ground is (g = 10 ms $^{-2}$)
Type	multiple choice
Option	2 km
Option	0.2 km
Option	20 km
Option	4 km
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Two bodies are projected horizontally at the same time from the top of a tower of height 78.4 m. If their velocities are 30 ms $^{-1}$ and 40 ms $^{-1}$ respectively. Find the difference between the times taken by them for hitting the ground.
Type	multiple choice
Option	2 sec
Option	4 sec
Option	1 sec
Option	Zero
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	A body is projected horizontally from the top of a tower with an initial velocity of 18 ms $^{-1}$. It hits the ground at an angle of 45° with horizontal. What is the vertical component of velocity when the body strikes the ground?
Type	multiple choice
Option	9 ms $^{-1}$
Option	$9\sqrt{2}$ ms $^{-1}$
Option	18 ms $^{-1}$
Option	$18\sqrt{2}$ ms $^{-1}$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A body of mass m thrown horizontally with velocity v, from the top of a tower of height h touches the level ground at a distance of 250 m from the foot of the

	tower. A body of mass 2m thrown horizontally with velocity $\frac{v}{2}$, from the top of a tower of height 4h will touch the level ground at a distance x from the foot of tower. The value of x is
Type	multiple choice
Option	250 m
Option	500 m
Option	125 m
Option	$250\sqrt{2}$ m
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	A projectile is fired horizontally with a speed of 98 ms^{-1} from the top of a hill 490 m high. Find the time taken to reach the ground
Type	multiple choice
Option	5 s,
Option	10 s,
Option	15 s,
Option	20 s.
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	A body is projected horizontally with a velocity of 10 ms^{-1} from the top of an inclined plane of angle 45°. If it is to hit the foot of the plane and acceleration due to gravity is 10 ms^{-2}, the height of the plane (in m) is
Type	multiple choice
Option	20
Option	40
Option	10
Option	15
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Two tall buildings face each other and are at a distance of $180 m$ from each other. With what velocity (in m/s) must a ball be thrown horizontally from a window $55 m$ above the ground in one building, so that it enters a window $10.9 m$ above the ground in the second building? (Take $g = 9.8 m/s^2$)
Type	multiple choice
Option	60
Option	40
Option	50
Option	30
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Find the speed of a projectile, fired horizontally from the top of a tower of height $80 m$ to reach the same distance as of the height of the tower.

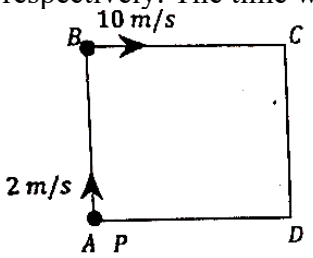
Type	multiple choice
Option	$20m/sec$
Option	$10m/sec$
Option	$30m/sec$
Option	$40m/sec$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	A train travels at $60m/s$ to the east with respect to the ground. A businessman on the train runs at $5m/s$ to the west with respect to the train. Find the velocity of the man with respect to the ground.
Type	multiple choice
Option	$5m/s$
Option	$55m/s$
Option	$60m/s$
Option	$65m/s$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	When two objects move uniformly towards each other, they get $4m$ closer in each second and when they move uniformly in the same direction with the original speeds, they get $4.0m$ closer in each $10s$. What are the speed of the objects?
Type	multiple choice
Option	$2.8m/s$ and $1.2m/s$
Option	$5.2m/s$ and $4.6m/s$
Option	$32m/s$ and $2.1m/s$
Option	$2.2m/s$ and $1.8m/s$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Two balls A and B are thrown vertically upward with velocity $5 m/s$ and $10 m/s$ respectively. Find separation between them after one second. (Take $g = 10m/s^2$)
Type	multiple choice
Option	$10 m$
Option	$5 m$
Option	$15 m$
Option	$0 m$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	One body is dropped, while a second body is thrown downward with an initial velocity of 1 m s^{-1} simultaneously. The separation between these is 18 m after a time
Type	multiple choice
Option	4.5 s
Option	5.5 s

Option	18 s
Option	36 s
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Two trains, each 50 m long are travelling in opposite direction with velocity 10 m/s and 15 m/s. The time of crossing is
Type	multiple choice
Option	2 s
Option	4 s
Option	$2\sqrt{3}s$
Option	$4\sqrt{3}s$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Particle A moves along X -axis with a uniform velocity of magnitude 10 m/s. Particle B moves with uniform velocity 20 m/s along a direction making an angle of 60° with the positive direction of X-axis as shown in figure. The relative velocity of B with respect to that of A is 
Type	multiple choice
Option	10 m/s along Y -axis
Option	$10\sqrt{3}\,m/s$ along $Y-$axis (perpendicular to $X-$axis)
Option	$10\sqrt{5}$ along the bisection of the velocity of A and B
Option	$30\,m/s$ along negative $X-$axis
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	A police jeep is chasing with velocity of 45 km/h a thief in another jeep moving with velocity 153 km/h. Police fires a bullet with muzzle velocity of 180 m/s. The velocity with which it will strike the car of the thief is
Type	multiple choice
Option	150 m/s
Option	27 m/s
Option	450 m/s
Option	250 m/s
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	A man is walking on horizontal road with a speed of 4 km/h. Suddenly, the rain starts vertically downwards with a speed of 7 km/h. The magnitude of relative velocity of the rain with respect the man is:
Type	multiple choice

Option	$\sqrt{33}\,km/h$
Option	$\sqrt{65}\,km/h$
Option	$8\,km/h$
Option	$4\,km/h$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	A man is walking on the road with a speed of 3m/s. Rain is falling vertically with a speed of 3 m/s. At what angle from the vertical man has to hold the umbrella to avoid the rain drops
Type	multiple choice
Option	45^0
Option	30^0
Option	60^0
Option	90^0
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	A person, reaches a point directly opposite on the other bank of a flowing river, while swimming at a speed of 5 m/s at an angle of 120° with the flow. The speed of the flow must be:
Type	multiple choice
Option	$2.5\,m/s$
Option	$3\,m/s$
Option	$4\,m/s$
Option	$1.5\,m/s$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	A man can row a boat with a speed of 10 kmph in still water. The river flows at 6 kmph. If he crosses the river from one bank to the other along the shortest possible path, time taken to cross the river of width 1 km is:
Type	multiple choice
Option	$1/8\,h$
Option	$1/4\,h$
Option	$1/2\,h$
Option	$1\,h$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	A man is crossing a river flowing with a velocity of 5 m/s. He reaches a point directly across the river at a distance of 60 m in 5s. His velocity in still water should be:
Type	multiple choice

Option	$12\,m/s$
Option	$13\,m/s$
Option	$5\,m/s$
Option	$10\,m/s$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	A car with a vertical windshield moves in a rain storm at a speed of. 40 km/hr. The rain drops fall vertically with a constant speed of 20 m/s. The angle at which raindrops strike the windshield is:
Type	multiple choice
Option	$\tan^{-1}\frac{5}{9}$
Option	$\tan^{-1}\frac{9}{5}$
Option	$\tan^{-1}\frac{3}{2}$
Option	$\tan^{-1}\frac{2}{3}$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	A river is flowing from W to E with a speed of 5 m/min. A man can swim in still water with a velocity of 10m/min. In which direction should the man swim so as to take the shortest possible path to go to the south.
Type	multiple choice
Option	30° with downstream
Option	60° with downstream
Option	120° with downstream
Option	South
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	It is raining at 20 m/s in still air. Now a wind starts blowing with speed 10 m/ sin the north direction. If a cyclist starts moving at 10 m/s in the south direction then the apparent velocity of rain with respect to a cyclist will be
Type	multiple choice
Option	$20\,m/s$
Option	$20\sqrt{2}\,m/s$
Option	$10\sqrt{5}\,m/s$
Option	$30\,m/s$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	A person, who can swim with speed u relative to water, wants to cross a river (of

	width d and water is flowing with speed v). The minimum time in which the person can do so is:
Type	multiple choice
Option	$\frac{d}{v}$
Option	$\frac{d}{u}$
Option	$\frac{d}{\sqrt{v^2+u^2}}$
Option	$\frac{d}{\sqrt{v^2-u^2}}$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	A boat which has a speed of 5 km/hr in still water crosses a river of width 1 km along the shortest possible path in 15 minutes. The velocity of the river water in km/hr is
Type	multiple choice
Option	3
Option	4
Option	1
Option	$\sqrt{21}$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	A car is moving along east at 10 m/s and a bus is moving along north at 10 m/s. The velocity of the car with respect to the bus is along:
Type	multiple choice
Option	North-East
Option	South-East
Option	North-West
Option	South-West
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Two bullets are fired horizontally and simultaneously towards each other from rooftops of two buildings (building being 100 m apart and being of the same height of 200 m)with the same velocity of 25 m/s. When and where will the two bullets collide? (g = 10 m/s^2)?
Type	multiple choice
Option	after 2s at a height of 180 m
Option	after 2s at a height of 20 m
Option	after 4s at a height of 120 m
Option	they will not collide
Answer	1
Solution	
Positive Marks	4
Negative Marks	1

Question	Two men P & Q are standing at corners A & B of square ABCD of side 8 m. They start moving along the track with constant speed 2m/s and 10 m/s respectively. The time when they will meet for the first time, is equal to: 
Type	multiple choice
Option	2 sec
Option	3 sec
Option	1 sec
Option	6 sec
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Two particles having position vectors $\vec{r}_1 = (3\hat{i} + 5\hat{j})$ metres and $\vec{r}_2 = (-5\hat{i} - 3\hat{j})$ meters are moving with velocity $\vec{v}_1 = (4\hat{i} + 3\hat{j})\text{m/s}$ and $\vec{v}_2 = (\alpha\hat{i} + 7\hat{j})\text{m/s}$. If they collide after 2 seconds, the value of α is
Type	multiple choice
Option	2
Option	4
Option	6
Option	8
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	A ship A is moving westwards with a speed of 10 kmph and a ship B 100 km south of A, is moving northwards with a speed of 10 kmph when the distance between them becomes the shortest is:
Type	multiple choice
Option	$0\,h$
Option	$5\,h$
Option	$5\sqrt{2}\,h$
Option	$10\sqrt{2}\,h$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1